

Cite this: *RSC Adv.*, 2016, 6, 83697

Mono-, bi- and multi-nuclear silver complexes constructed from bis(benzimidazole)-2-oxapropane ligands and methacrylate: syntheses, crystal structures, DNA-binding properties and antioxidant activities†

Han Zhang, Yuling Xu, Huilu Wu,* Stephen Opeyemi Aderinto and Xuyang Fan

The reactions of AgNO_3 with sodium methacrylate and three structurally related ligands in the absence of light gave three new silver coordination complexes, namely, $[\text{Ag}_2(\text{methacrylate})_2(\text{Etobb})_2] \cdot \text{CH}_3\text{CN}$ **1**, $[\text{Ag}(\text{methacrylate})(\text{Bobb})]$ **2** and $[\text{Ag}_2(\text{methacrylate})_2(\text{Aobb})]_n$ **3**, where Etobb = 1,3-bis(1-ethylbenzimidazol-2-yl)-2-oxapropane, Bobb = 1,3-bis(1-benzylbenzimidazol-2-yl)-2-oxapropane, Aobb = 1,3-bis(1-allylbenzimidazol-2-yl)-2-oxapropane. All three complexes have been characterized by elemental analysis, UV-visible and infrared spectroscopy, and single-crystal X-ray diffraction. Complex **1** is binuclear and tri-coordinated by two N atoms to two Etobb ligands, while complex **2** is mononuclear. Complex **3** is a unique metal-organic compound that has a diamond-like multinuclear silver center, with each Ag bridged by two Aobb ligands and two methacrylate ions to form 1-D single-coordination polymer chain structures that extend into 2-D frameworks through π - π interactions. The X-ray structures show a wide range of Ag-Ag distances in the polymers, which are dependent on the conformation of the bridging ligand. The DNA-binding properties of the three Ag(I) complexes have been investigated by electronic absorption, fluorescence, and viscosity measurements. The results suggest that the three complexes bind to DNA via an intercalative mode, and their binding affinity for DNA follows the order $1 > 2 > 3$. Moreover, the antioxidant activity experiment shows that complex **2** has the potential ability to scavenge hydroxyl radicals in *in vitro* studies.

Received 15th April 2016

Accepted 31st July 2016

DOI: 10.1039/c6ra09733b

www.rsc.org/advances

Introduction

Medicinal inorganic chemistry offers possibilities for the design of therapeutic agents not readily available to organic compounds.^{1,2} The wide range of coordination numbers and geometries, accessible redox states, thermodynamic and kinetic characteristics, and the intrinsic properties of the cationic metal ion and the ligand itself offer to the medicinal chemist a wide spectrum of reactivities that can be exploited.³ A number of Cu(I), Au(I) and Ag(I) phosphane complexes have been investigated for their antitumor properties,^{4–8} making possible the assessment of a structure-activity relationship (SAR) for a wide range of metal complexes.

Benzimidazole derivatives are important constituents in most pharmacologically, catalytically, and biologically active

compounds, and they therefore correspond to significant synthetic targets.^{9–11} Ag(I) is a well-known, versatile ion that is often used in the construction of coordination clusters because of its coordination diversity as well as its positive coordination tendency with various donor atoms.^{12–14} The compounds formed by the Ag(I) atom and multi-topic N-donor ligands have attracted great interest for a number of reasons. The soft d^{10} Ag^+ ion has been commonly used as a metal center in the construction of coordination polymers.¹⁵ The Ag(I) center has a closed-shell electronic configuration and can adopt diverse coordination numbers from 2 to 8, with no strong energetic preference for any particular geometry,^{16,17} varying from linear to trigonal, tetragonal, square pyramidal, octahedral, and so on. In recent years, researchers have explored silver, gold and platinum-based NHCs with benzimidazole cores as excellent antitumor and anticancer agents in the form of complexes. Their various biological applications with fewer side effects are now attracting worldwide attention.¹⁸

In this paper, three new silver coordination complexes have been synthesized through the variation of the structurally-different N-donor ligands. In addition, the DNA-binding properties and antioxidant activities of the coordination compounds have been studied.

School of Chemical and Biological Engineering, Lanzhou Jiaotong University, Lanzhou, Gansu, 730070, PR China. E-mail: wuhuilu@163.com

† CCDC Crystallographic data for complexes **1–3** have been deposited with the Cambridge Crystallographic Data Center as supplementary publication CCDC 996125, 996126 and 1012704. For crystallographic data in CIF or other electronic format see DOI: 10.1039/c6ra09733b

Results and discussion

Characterization of the complexes

The free ligands Etobb (1,3-bis(1-ethylbenzimidazol-2-yl)-2-oxapropane), Bobb (1,3-bis(1-benzylbenzimidazol-2-yl)-2-oxapropane), Aobb (1,3-bis(1-allylbenzimidazol-2-yl)-2-oxapropane) and Ag(I) complexes are stable in both air and solution environments. The syntheses of complexes **1–3** were carried out in the dark to avoid photodecomposition. These free ligands are soluble in organic solvents but insoluble in water. The Ag(I) complexes are soluble in DMF and DMSO, and partially soluble in water and other organic solvents such as methanol, ethanol, acetone, petroleum ether, trichloromethane, *etc.* The results of the elemental analyses show that the composition of complex **1** is $[\text{Ag}_2(\text{methacrylate})_2(\text{Etobb})_2] \cdot \text{CH}_3\text{CN}$, that of complex **2** is $[\text{Ag}(\text{methacrylate})(\text{Bobb})]$, and that of complex **3** is $[\text{Ag}_2(\text{methacrylate})_2(\text{Aobb})]_n$. The molar conductance values show that the three compounds are non-electrolytes in DMF.¹⁹

The IR spectra of the free ligand Etobb and its Ag(I) complex **1** were compared. Free Etobb exhibits a characteristic C=N stretching frequency at 1475 cm^{-1} , whereas the C=N stretching frequency of the complex is observed at 1552 cm^{-1} . Hence, the C=N stretching frequency is shifted upon complexation,²⁰ indicating that the nitrogen atoms of the ligand are coordinated to the Ag(I) center. Similar shifts also appeared in complexes **2** and **3**, which led to the same conclusion. Free Bobb and Aobb exhibit characteristic C=N stretching frequencies at 1496 and 1463 cm^{-1} , whereas the C=N stretching frequencies of complexes **2** and **3** are observed at 1552 and 1558 cm^{-1} . Our deduction agrees with the results of the X-ray crystal structure determinations.

As expected, DMF solutions of the ligands and Ag(I) complexes show almost identical UV spectra. From the UV-vis spectra, the band observed for the free Etobb is marginally red-shifted by 1 nm in complex **1**, again confirming the coordination of the C=N group to the Ag(I) center. The absorption bands at 280 nm are assigned to $\pi\text{--}\pi^*$ (imidazole) transitions.²¹ Analogously, the UV bands of Bobb (279, 287 nm) and Aobb (279, 287 nm) are also marginally red-shifted, by about 1–2 nm, in both complexes **2** and **3**. This phenomenon further shows that C=N is involved in coordination to the metal center.

X-ray structures of the complexes

The structures of **1**, **2** and **3** were confirmed by X-ray diffraction. The crystal structures are shown in Fig. 1, 3 and 5, respectively.

Structure of complex 1. Crystallographic analysis reveals that complex **1** crystallizes in the monoclinic space group $P2(1)/n$ and that the asymmetric unit of complex **1** has two Ag(I) atoms, two Etobb ligands and two methacrylate anions, as shown in Fig. 1. In complex **1**, Ag1 and Ag2 have the same coordination environments, which are binuclear and tri-coordinated by one oxygen atom from one methacrylate anion [$\text{Ag1--O2} = 2.493(2)\text{ \AA}$] and two nitrogen atoms from two Etobb ligands [$\text{Ag1--N4} = 2.2268(19)$, $\text{Ag1--N2} = 2.2476(19)\text{ \AA}$]. The N2–Ag1–N4 angle is reduced to $142.58(7)^\circ$, compared to the somewhat larger angle in the previously-reported Ag complexes with similar ligand architectures.²² The interaction bond length between Ag(I) atom and oxygen is $2.608\text{ \AA}$, which is consistent with those of similar isoelectronic complexes reported previously.^{23,24} However, the interaction distance is shorter than the sum of van der Waals radii ($3.24\text{ \AA}$) of the respective elements for an Ag–O interaction.²⁵ Probably, this insight may be useful in determining the

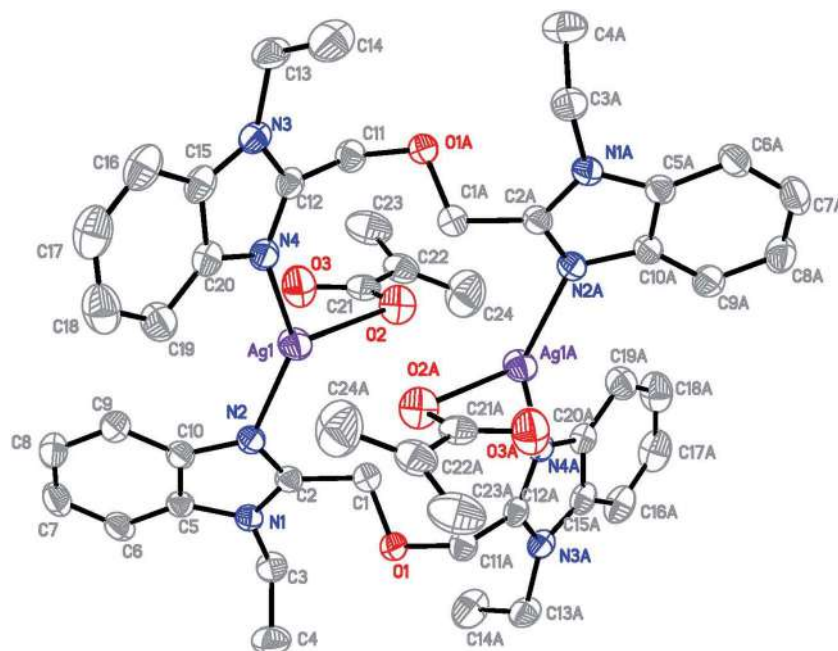


Fig. 1 Molecular structure of complex **1** with displacement ellipsoids drawn at the 30% probability level; solvent molecules and H atoms are omitted for clarity.

mode of action of complex **1** as anticancer agent, as it may form a strong interaction with the DNA of an infected cell. In the extended structure, the complex **1** unit extends to 1-D supramolecular chains through C–H(benzimidazole)⋯O(methacrylate) with bond distances ranging from 2.493 to 3.314 Å, and it is further consolidated by π – π stacking interactions with bond distances of 3.604 Å, as shown in Fig. 2.

Structure of complex 2. The X-ray structural characterization shows that complex **2** belongs to the triclinic crystal system, crystallizing in the space group $P\bar{1}$. Complex **2** is a mononuclear, tri-coordinating configuration, with the structure consisting of a central metal Ag(I) atom, a Bobb and a methacrylate (Fig. 3). The Ag(I) atom displays a trigonal planar coordination structure consisting of one oxygen atom from one methacrylate anion and two nitrogen atoms from one Bobb ligand. The Ag–O bond distance is 2.313(2) Å, falling into the normal ranges.²⁶ Similarly, the complex **2** unit extends to a 1-D supramolecular chain

through weak hydrogen C–H(benzyl)⋯O(methacrylate) and Ag⋯H–C, as shown in Fig. 4.

More interestingly, the intramolecular distances further show the presence of obvious C–H⋯M close intramolecular interactions between the Ag center and the Bobb H(1B) atom, with Ag⋯H distances of 2.811 Å, Ag⋯C distances of 3.739 Å ($d(\text{Ag}\cdots\text{H}) < d(\text{Ag}\cdots\text{C})$), and C–H⋯Ag angles of 160.362° (C–H⋯Ag $> 100^\circ$), which, to the best of our knowledge, are similar to the corresponding values for reported Ag(I) complexes.²⁷ Therefore, these C–H⋯M close interactions should be correctly described as being weak intramolecular C–H⋯M hydrogen bonding (a special case of hydrogen bonding); this description is also chemically and geometrically more reasonable.^{28–30} Moreover, from the viewpoint of geometrical requirements, the C–H⋯Ag weak interactions may have facilitated the formation of the coordination bond between Ag(I) and the N donor of the benzimidazole ring in the solid state.

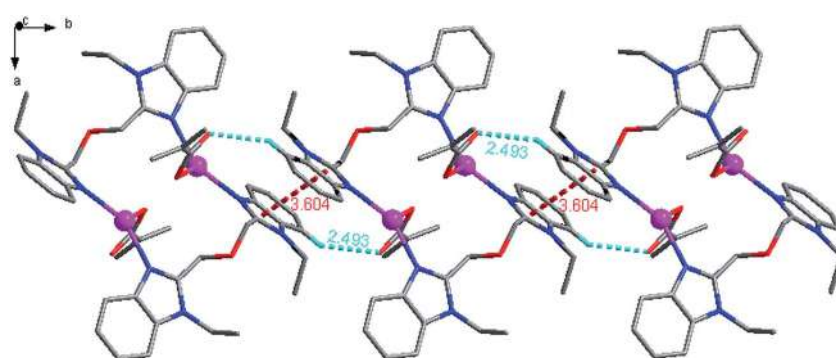


Fig. 2 The structure of complex **1** linked via π – π stacking and weak hydrogen interactions.

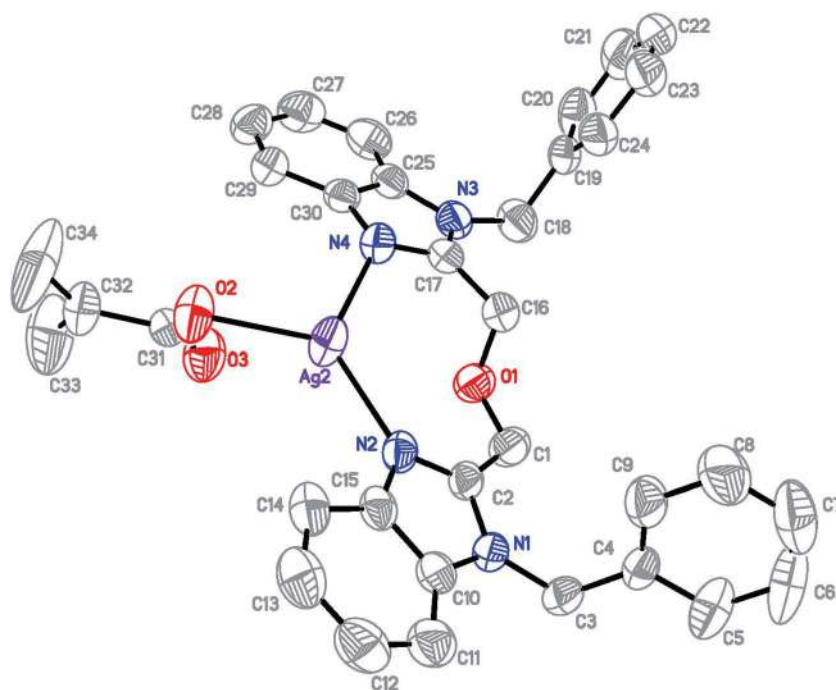


Fig. 3 The coordination environment of the Ag(I) cation in **2** showing 30% thermal probability ellipsoids. The H atoms are omitted for clarity.

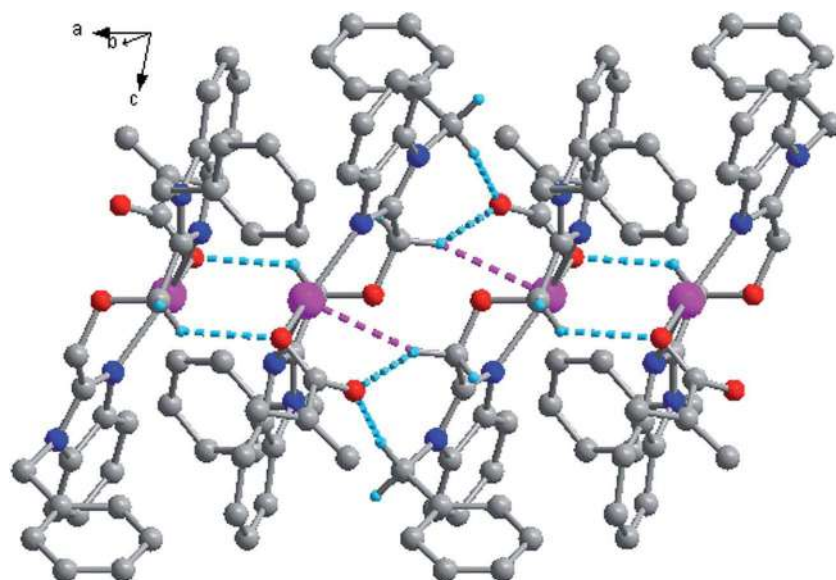


Fig. 4 The 1D supramolecular chain of **2** formed with C–H···O hydrogen bonding and C–H···Ag H-bonding interactions.

Structure of complex 3. Single-crystal X-ray diffraction analysis reveals that complex **3** crystallizes in the monoclinic space group $C2/c$. Complex **3** is a unique metal organic compound and has a diamond-like multinuclear silver center with each Ag(I) atom bridged by two Aobb and two methacrylate ions to form 1D single-coordination polymer chain structures. Its unit

possesses two Ag(I) atoms, one Aobb ligand and two methacrylate anions, where the component moieties all lie in general positions with no crystallographically imposed symmetry (Fig. 5). Each of the Ag(I) atoms has the same five-coordinated environment, which exhibits a distorted trigonal bipyramidal structure comprising two oxygen atoms from two methacrylate

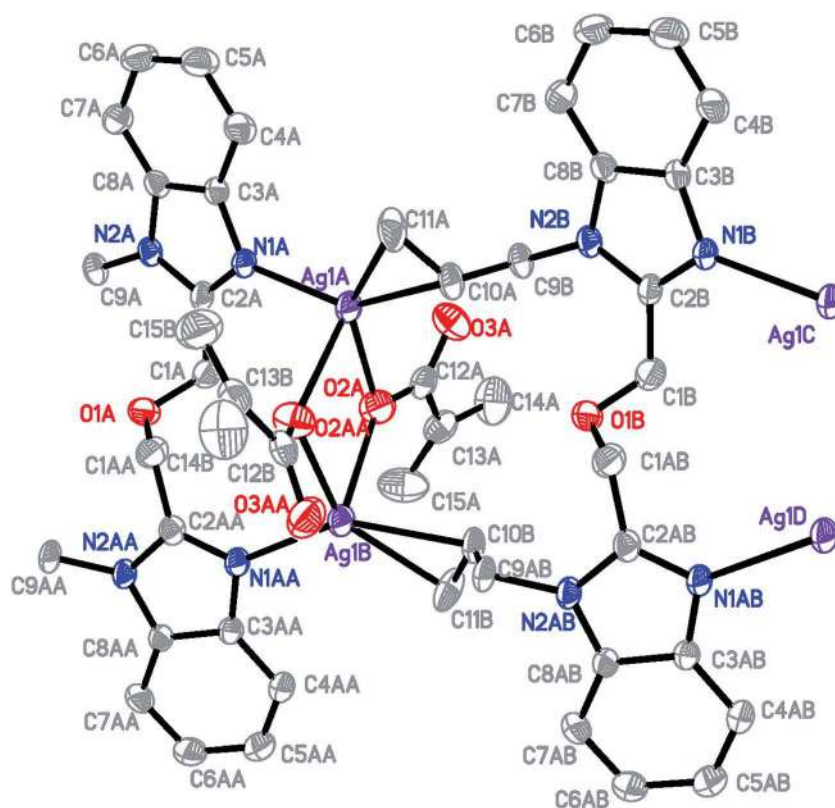


Fig. 5 The environment of the Ag(I) cation in **3** showing 30% thermal probability ellipsoids. The H atoms are omitted for clarity.

anions [Ag–O = 2.309(2) and 2.634(2) Å], one nitrogen atom from one Aobb ligand [Ag–N = 2.270(2) Å], and two carbon atoms from one allyl(Aobb) [C(10)–Ag(1) = 2.388(2) Å and C(11)–Ag(1) = 2.334(3) Å]. As shown in Fig. 6, unlike the structures of complexes **1** and **2**, in complex **3** the ligand Aobb is a bridging ligand. It does not only act as a σ -bonding ligand but also as a π -acid ligand. The adjacent Ag(I) atoms and the allyl substituents on the Aobb bridge to form a one-dimensional supramolecular polymer chain ligand. The two Ag–C distances [C(10)–Ag(1) = 2.388(2) Å and C(11)–Ag(1) = 2.334(3) Å] are comfortably within the sum of the van der Waals radius of allyl and the ionic radius of Ag(I). The 1-D single-polymer chain is extended into 2-D frameworks through strong π – π interactions (Fig. 7).

DNA-binding properties

Electronic absorption titration. We first investigated the binding modes of DNA with three Ag(I) complexes through absorption titration experiments. The electronic absorption spectra of the three Ag(I) complexes in the absence and presence of CT-DNA are given in Fig. 8. As seen in Fig. 6a–c, there is a band at 270 nm for the complexes. Upon the addition of increasing CT-DNA concentrations, the band exhibited hypochromism of 58.6%, 49.1% and 17.9%, respectively. A compound binding to DNA through intercalation usually shows

hypochromism due to the intercalation mode involving a strong π – π stacking interaction between the aromatic chromophore and the DNA base pairs. It is generally accepted that the extent of hypochromism in the UV-vis band is consistent with the strength of intercalation.³¹ Hence, these observations suggest that the compounds interact with CT-DNA by intercalation. From the electronic absorption spectroscopy experiments, K_b values of complexes **1**–**3** are obtained and shown in Table 1. The present K_b values are comparable with the values for the so-called DNA-intercalative complexes that have been reported in the literature ($K_b = 2.9 \times 10^4$ – $1.2 \times 10^5 \text{ M}^{-1}$).^{32–36} Furthermore, they also indicate that the binding affinities of the Ag(I) complexes follow the order: **1** > **2** > **3**. This may be attributed to three possible reasons: (i) the charge transfer of coordinated ligands caused by the coordination of the silver atom results in the decline of charge density of the plane conjugate system, which contributes to intercalation.^{37–39} (ii) By comparing their molecular structures, we find a greater number of coplanar aromatic rings, which contribute to intercalation with the base pairs of double-helical DNA and may result in stronger affinity for DNA.^{22,40,41} (iii) The reason for the differences in binding strengths among the three Ag(I) complexes can also be attributed to the differences in steric hindrance, which are caused by both the introduction of substituents and the geometric structure. The C1–O1–C11A and C1A–O1A–C11 angles in complex **1**

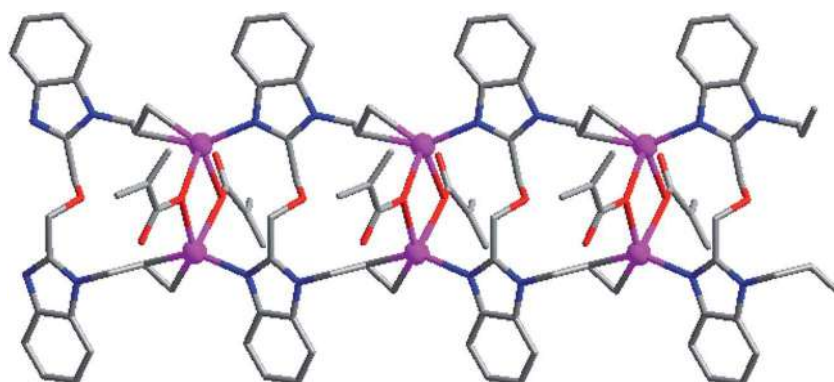


Fig. 6 The polymeric chain structure of complex **3**.

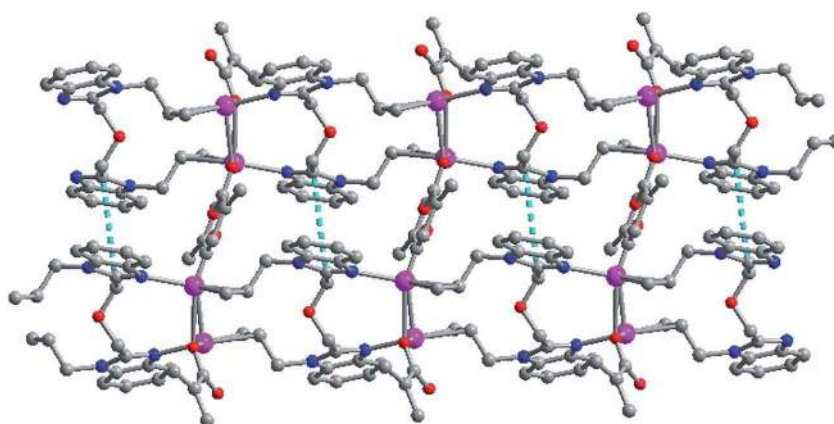


Fig. 7 An infinite 2-D supramolecular layer formed via π – π interactions in **3** (π – π interactions: dashed lines).

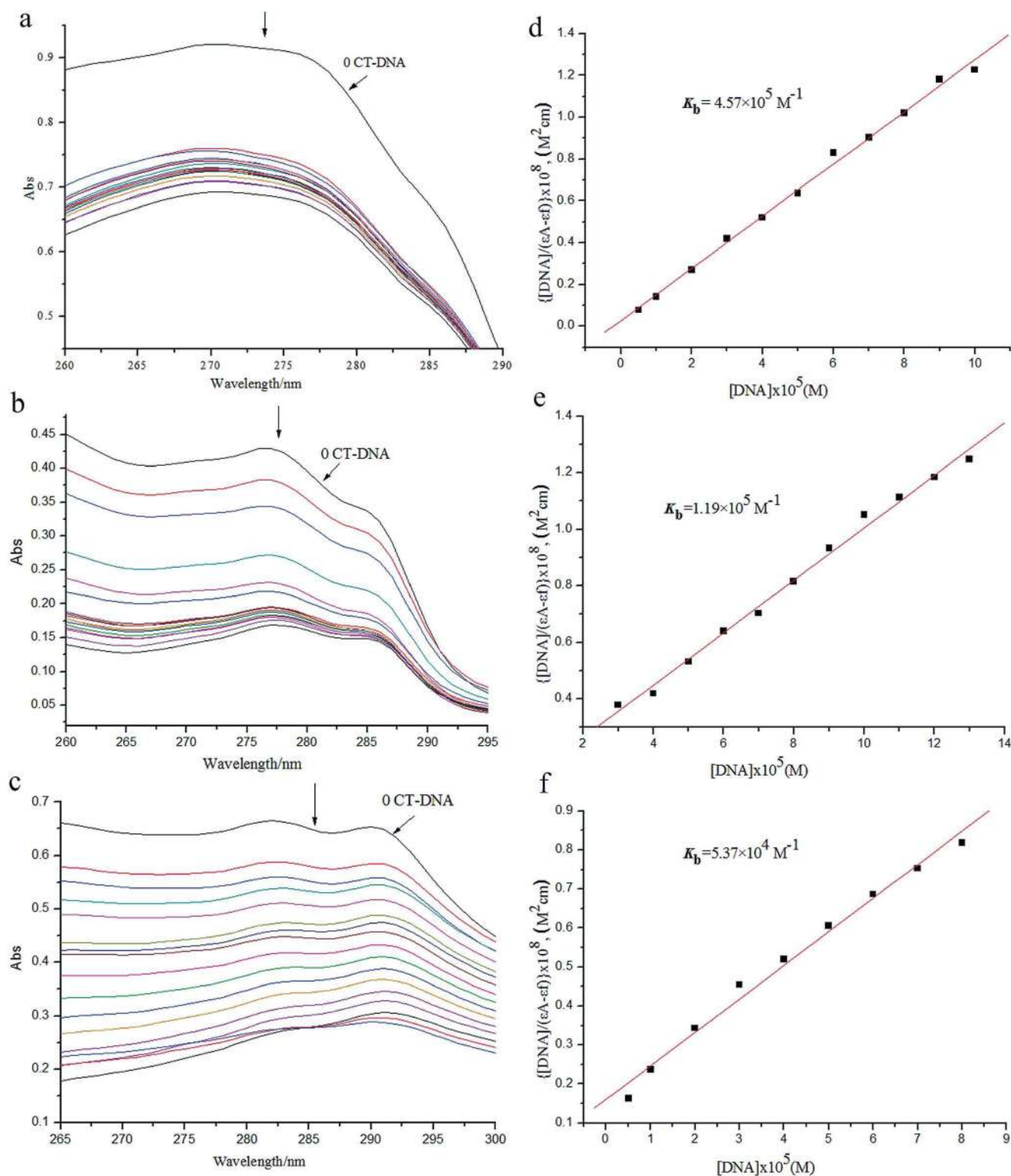


Fig. 8 Absorption spectra of complexes 1–3 (a–c) in Tris–HCl buffer upon addition of CT-DNA. The arrow shows the emission intensity changes with increasing DNA concentrations. $[DNA]/(\epsilon_A - \epsilon_f)$ versus $[DNA]$ for the titration of complexes 1–3 (d–f) with CT-DNA.

are 112.44° , whereas the C1–O1–C16 and C8–O3–C8A angles in complexes 2 and 3 are 113.12° and 114.44° , respectively, thus providing further support for this interpretation. These results further illustrate that electron density and steric hindrance play

a very important role in the DNA binding of the three Ag(I) complexes.

Fluorescence spectra. We then investigated the DNA binding by fluorescence spectroscopy. Three Ag(I) complexes show no

Table 1 Hypochromism, K_b and K_{sv} values for complexes 1–3

Complex	Hypochromism	k_b (M^{-1})	k_{sv} (M^{-1})
1	58.6%	4.57×10^5	1.68×10^3
2	49.1%	1.19×10^5	1.42×10^3
3	17.9%	5.37×10^4	1.25×10^3

fluorescence at room temperature in solution or in the presence of CT-DNA. However, competitive ethidium bromide (EB) binding studies could be undertaken to examine the binding of each compound with DNA. EB did not show any appreciable emission in buffer solution due to fluorescence quenching by the solvent. Upon the addition of the complex to a solution

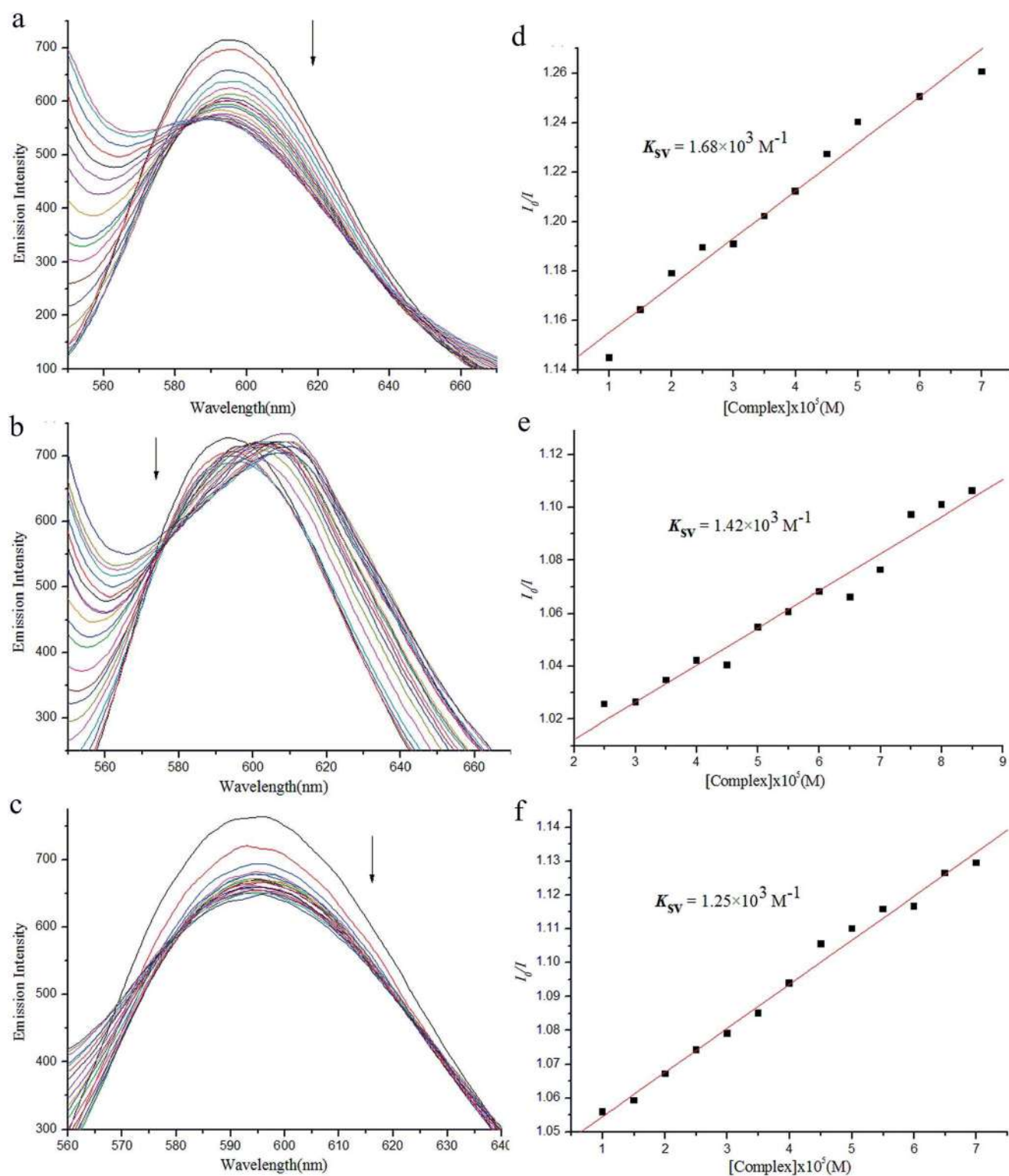


Fig. 9 Emission spectra of EB bound to CT-DNA in the presence of complexes 1–3 (a–c), $\lambda_{ex} = 520$ (nm). The arrows show the intensity changes upon increasing concentration of the complexes. The Stern–Volmer quenching plots of the complexes 1–3 (d–f) with increasing concentrations of CT-DNA, t .

containing EB, no change in the fluorescence spectra was observed. However, the fluorescence intensity of EB was greatly enhanced upon the addition of CT-DNA, due to its strong intercalation with the DNA base pairs. The addition of a second molecule, which may bind to DNA more strongly than EB, can result in a decreased DNA-induced EB emission by the displacement of EB.⁴² The emission spectra of EB bound to CT-DNA in the absence and presence of increasing amounts of each of the compounds were recorded. The addition of the compounds results in a significant decrease in emission band intensity of the DNA-EB system at 593 nm, indicating a competitive binding of the compounds to DNA. The Stern–Volmer plots of DNA-EB (Fig. 9) show that the quenching of EB bound to DNA with the three complexes follows a linear relationship, consistent with intercalation of the test compounds. The K_{sv} values for complexes 1–3 are given in Table 1. Moreover, the binding strengths of the complexes follow the same order as that obtained from the UV-vis experiments. In regard to this, relevant literature reports that silver ion binds to DNA.^{43–45} However, in the present study, rather than binding of the silver ion to DNA, the Ag complexes bind to DNA, and this occurs *via* intercalation modes due to the conjugate effect and hydrogen-bonding interaction with the ligands of bis(benzimidazole)-2-oxapropane. The conjugate effect and hydrogen bonding are physical bonds and thus will not alter or destroy DNA conformation.

Viscosity measurements. Finally, we used viscosity measurements to probe the interactions of these compounds with DNA. In the absence of crystallographic structural data, hydrodynamic measurements that are sensitive to DNA length changes are regarded as the least ambiguous and most critical tests of a binding model in solution.⁴⁶ For the Ag(I) complexes, as increasing amounts of the compounds were added, the viscosity of DNA increased steadily. The values of $(\eta/\eta_0)^{1/3}$ were plotted against [compound]/[DNA] (Fig. 10). In classical intercalation, the DNA helix lengthens as the base pairs are separated to accommodate the bound ligand, leading to increased

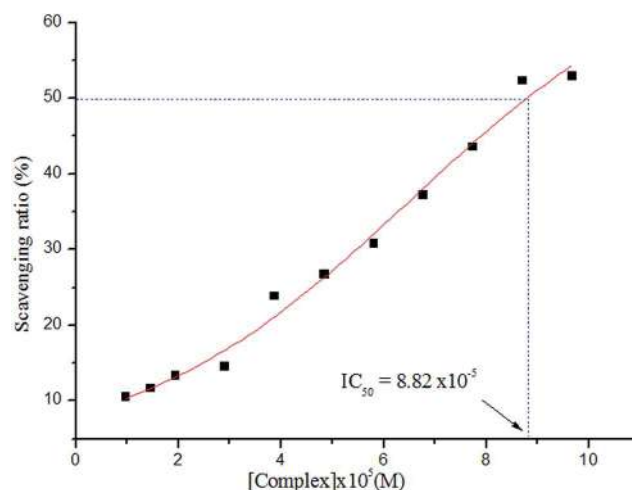


Fig. 11 Plots of hydroxyl radical scavenging effect (%) for complex 2.

DNA viscosity, whereas a partial, non-classical ligand intercalation causes a bend (or kink) in the DNA helix, reducing its effective length and thereby its viscosity.⁴⁷ Hence, these results provide further evidence that the complexes intercalate with CT-DNA, and DNA binding affinities follow the order: 1 > 2 > 3, which is consistent with the above absorption and fluorescence spectral results.

Hydroxyl radical scavenging activity. Fig. 11 depicts the inhibitory effect of the compounds on OH[•] radicals. The inhibitory activity of the compounds is marked, and the suppression ratio increases with increasing concentration of the test compound. We compared the ability of the present compound to scavenge hydroxyl radicals with those of the well-known natural antioxidant mannitol, using the same method as reported in previous papers.^{48,49} The 50% inhibitory concentration (IC_{50}) values of mannitol and vitamin C are about 9.6×10^{-3} and $8.7 \times 10^{-3} \text{ M}^{-1}$, respectively. According to the antioxidant experiments, the IC_{50} values of the Ag(I) complex 2 is $8.82 \times 10^{-5} \text{ M}$, which implies that complex 2 exhibits better scavenging activity than mannitol and vitamin C. Based on the observed IC_{50} values, complex 2 can be considered as a potential drug to eliminate the hydroxyl radical (OH[•]).

Conclusion

In conclusion, three mono-, bi- and multi-nuclear silver complexes with different flexible bis(benzimidazol-2-yl)-2-oxapropane ligands and the same methacrylate anion have been synthesized and structurally characterized. Complex 3 exhibited a distinct structure from those of complexes 1 and 2, which is a unique coordination polymer. These compounds display diverse architectures, from 1-D to 2-D. The results indicated that the flexible bis(benzimidazole)-2-oxapropane ligands and methacrylate ion have important effects on geometric structures of the silver complexes. DNA binding of the Ag(I) complexes suggests that the three Ag(I) complexes bind to DNA through an intercalation mode owing to the large planar aromatic rings, hydrogen bonds and π - π stacking interactions

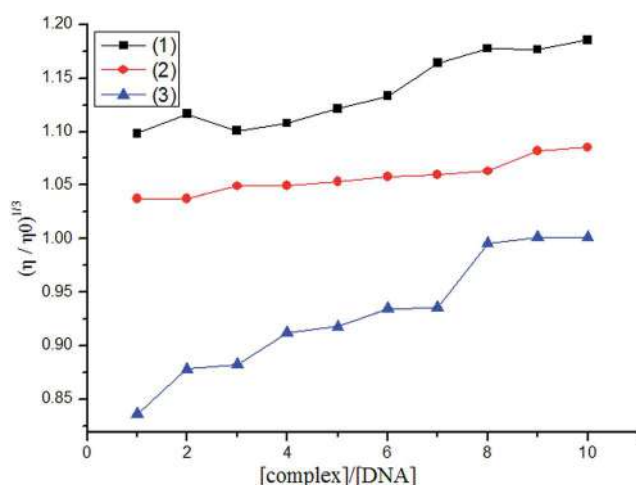


Fig. 10 Effect of increasing amounts of complexes 1–3 on the relative viscosity of DNA at 25.0 ± 0.1 °C.

that facilitate their intercalation into the base pairs of double-helical DNA. The DNA-binding affinities of the three complexes follow the order $1 > 2 > 3$. This result illustrates that steric hindrance and charge effects play a vital role in the DNA binding of the three Ag(I) complexes. Moreover, complex 2 has hydroxyl radical scavenging activity. These results should demonstrate the usefulness of Ag(I) complexes in the design of new functional materials with an impressive range of potential applications, including new antitumor drugs and antioxidants.

Experimental section

Materials and measurements

^1H NMR spectra were recorded on a Varian VR400 MHz spectrometer with TMS as an internal standard. Melting points were detected on an X-4 digital micro melting-point apparatus. Electrolytic conductance measurements were made with a DDS-11A-type conductivity bridge using 10^{-3} mol L^{-1} solutions in DMF at room temperature. IR spectra were recorded in the $4000\text{--}400$ cm^{-1} region with a Nicolet FT-VERTEX 70 spectrometer using KBr pellets. Electronic spectra were recorded on a Lab-Tech UV Bluestar spectrophotometer. C, H and N elemental analyses were determined using a Carlo Erba 1106 elemental analyzer. The fluorescence spectra were recorded on a Perkin Elmer LS-55 spectrofluorophotometer. The antioxidant activities against the hydroxyl and superoxide anion radicals were measured in a water bath with a Spectrumbiol 722sp spectrophotometer.

Calf thymus DNA (CT-DNA) and ethidium bromide (EB) were obtained from Sigma-Aldrich. All chemicals used for experiments were of analytical grade. Other reagents and solvents were reagent grade, obtained from commercial sources and used without further purification. Tris-HCl buffer, $\text{K}_2\text{HPO}_4\text{--}\text{KH}_2\text{PO}_4$ buffer, and EDTA-Fe(II) solution were prepared using doubly distilled water. Stock solutions of the complexes were prepared in DMF at 3×10^{-3} mol L^{-1} . The experiments involving interaction of the ligand and the complexes with CT-DNA were carried out in doubly distilled buffered water containing 5 mM Tris and 50 mM NaCl, adjusted to pH 7.2 with hydrochloric acid. A solution of CT-DNA gave a ratio of UV absorbance at 260 and 280 nm of about 1.8–1.9, indicating that the CT-DNA was sufficiently free of protein.⁵⁰ The CT-DNA concentration per nucleotide was determined spectrophotometrically by employing an extinction coefficient of $6600 \text{ M}^{-1} \text{ cm}^{-1}$ at 260 nm.⁵¹

Electronic absorption titration

Absorption titration experiments were performed with fixed concentrations of the complex while gradually increasing the concentration of CT-DNA. To obtain the absorption spectra, the required amount of CT-DNA was added to both the compound and reference solutions to eliminate the absorbance of CT-DNA itself. From the absorption titration data, the binding constant (K_b) was determined using the following equation:⁵² $[\text{DNA}]/(\epsilon_a - \epsilon_f) = [\text{DNA}]/(\epsilon_b - \epsilon_f) + 1/K_b(\epsilon_b - \epsilon_f)$, where $[\text{DNA}]$ is the concentration of CT-DNA in base pairs, ϵ_a corresponds to the

observed extinction coefficient ($A_{\text{obsd}}/[\text{M}]$), ϵ_f corresponds to the extinction coefficient of the free compound, ϵ_b is the extinction coefficient of the complex when fully bound to CT-DNA, and K_b is the intrinsic binding constant. The ratio of slope to intercept in the plot of $[\text{DNA}]/(\epsilon_a - \epsilon_f)$ versus $[\text{DNA}]$ gave the value of K_b .

Competitive binding with ethidium bromide

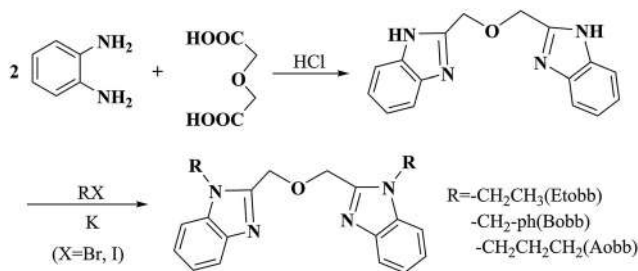
The extent of fluorescence quenching of EB bound to CT-DNA can be used to determine the extent of binding between an additional molecule and CT-DNA.⁵³ The competitive binding experiments were carried out in the buffer by keeping $[\text{DNA}]/[\text{EB}] = 1.13$ and varying the concentrations of the compounds. The fluorescence spectra of EB were measured using an excitation wavelength of 520 nm, and the emission range was set between 550 and 750 nm. The spectra were analyzed according to the classical Stern-Volmer equation:⁵⁴ $I_0/I = 1 + K_{\text{sv}}[Q]$, where I_0 and I are the fluorescence intensities at 599 nm in the absence and presence of the quencher, respectively, K_{sv} is the linear Stern-Volmer quenching constant, and $[Q]$ is the concentration of the quencher. In these experiments, $[\text{CT-DNA}] = 2.5 \times 10^{-3}$ mol L^{-1} , and $[\text{EB}] = 2.2 \times 10^{-3}$ mol L^{-1} .

Viscosity titration measurements

Viscosity experiments were conducted on an Ubbelodhe viscometer immersed in a water bath maintained at 25.0 ± 0.1 °C. Titrations were performed for the complex (3 μM), and each compound was introduced into the CT-DNA solution (50 μM) present in the viscometer. Data were analyzed as $(\eta/\eta_0)^{1/3}$ versus the ratio of the concentration of the compound to CT-DNA, where η is the viscosity of CT-DNA in the presence of the compound and η_0 is the viscosity of CT-DNA alone. Viscosity values were calculated from the observed flow times of CT-DNA containing solutions corrected for the flow time of buffer alone (t_0), $\eta = (t - t_0)$.⁵⁵

Hydroxyl radical scavenger measurements

Hydroxyl radicals were generated in aqueous media through a Fenton-type reaction. The reaction mixtures (3 mL) contained 1 mL of 40 $\mu\text{g mL}^{-1}$ aqueous safranin, 1 mL of 1.0 mmol aqueous EDTA-Fe(II), 1 mL of 3% aqueous H_2O_2 , and a series of quantitative micro-additions of the test compound. A sample without the test compound was used as the control. The



Scheme 1 Synthetic route for ligands (Etobb, Bobb, Aobb).

reaction mixtures were incubated at 37 °C for 30 min in a water bath. The absorbance at 520 nm was then measured to correct for the solvent effect. The scavenging effect for OH[•] was calculated from the following expression: scavenging effect % = $(A_{\text{sample}} - A_r)/(A_o - A_r) \times 100\%$, where A_{sample} is the absorbance of the sample in the presence of the test compound, A_r is the

absorbance of the blank in the absence of the test compound, and A_o is the absorbance in the absence of the test compound and EDTA-Fe(II).⁵⁶

Synthesis of Etohb, Bobb and Aobb. These ligands were synthesized according to the literature methods,^{57–59} as shown in Scheme 1, with the following results.

Table 2 Crystal data and structure refinement for complexes 1–3

Complex	1	2	3
Empirical formula	C ₅₂ H ₆₀ Ag ₂ N ₁₀ O ₆	C ₃₄ H ₃₁ AgN ₄ O ₃	C ₃₀ H ₃₂ Ag ₂ N ₄ O ₅
Formula weight	1136.84	651.50	744.34
Crystal system	Monoclinic	Triclinic	Monoclinic
Space group	<i>P</i> 2(1)/ <i>n</i>	<i>P</i> $\bar{1}$	<i>C</i> 2/ <i>c</i>
<i>a</i> [Å]	11.555(4)	9.0046(4)	18.7281(16)
<i>b</i> [Å]	10.932(4)	12.9691(9)	8.7563(7)
<i>c</i> [Å]	20.204(7)	13.8963(9)	17.4903(15)
α [°]	90	70.317(6)	90
β [°]	92.210(3)	79.808(5)	92.3340(10)
γ [°]	90	73.946(5)	90
<i>V</i> [Å ³]	2550.1(14)	1462.11(15)	2865.8(4)
<i>Z</i>	2	2	4
ρ_{calcd} [mg m ^{−3}]	1.481	1.480	1.725
μ [mm ^{−1}]	0.827	0.731	1.413
<i>F</i> (000)	1168	668	1496
<i>h</i> / <i>k</i> / <i>l</i> [max, min]	−13,13/−11,13/−24,24	−10,10/−15,15/−16,10	−22,22/−10,10/−21,15
θ Range for data collection [°]	2.00 to 25.50	2.95 to 25.00	2.18 to 25.50
Goodness-of-fit on <i>F</i> ²	1.071	1.050	1.112
Final <i>R</i> ₁ , <i>wR</i> ₂ indices [<i>I</i> > 2σ(<i>I</i>)]	0.0280/0.0775	0.0365/0.0736	0.0248/0.0636
<i>R</i> ₁ , <i>wR</i> ₂ indices [all data]	0.0316/0.0799	0.0489/0.0813	0.0267/0.0690

Table 3 Selected bond lengths (Å) and bond angles (°) of complexes 1–3^a

Complex 1				
Bond distances	Ag(1)–N(4)	2.2268(19)	Ag(1)–N(2)	2.2476(19)
	Ag(1)–O(2)	2.493(2)		
Bond angles	N(4)–Ag(1)–N(2)	142.58(7)	N(4)–Ag(1)–O(2)	110.78(7)
	N(2)–Ag(1)–O(2)	104.35(7)	C(31)–O(2)–Ag(2)	102.2(2)
	C(17)–N(4)–Ag(2)	116.3(2)	C(30)–N(4)–Ag(2)	131.04(19)
	C(2)–N(2)–Ag(2)	133.0(2)	C(15)–N(2)–Ag(2)	121.62(19)
Complex 2				
Bond distances	Ag(2)–N(2)	2.245(2)	Ag(2)–O(2)	2.313(2)
	Ag(2)–N(4)	2.336(3)		
Bond angles	N(2)–Ag(2)–O(2)	129.19(9)	N(2)–Ag(2)–N(4)	121.89(9)
	O(2)–Ag(2)–N(4)	108.50(9)	C(31)–O(2)–Ag(2)	102.2(2)
	C(17)–N(4)–Ag(2)	116.3(2)	C(30)–N(4)–Ag(2)	131.04(19)
	C(2)–N(2)–Ag(2)	133.0(2)	C(15)–N(2)–Ag(2)	121.62(19)
Complex 3				
Bond distances	Ag(1)–N(1)	2.270(2)	Ag(1A)–O(2A)	2.309(2)
	Ag(1A)–O(2AA)	2.634(2)	Ag(1B)–O(2AA)	2.309(2)
	Ag(1B)–O(2A)	2.634(2)	Ag(1)–C(10)#1	2.388(2)
	Ag(1)–C(11)#1	2.334(3)		
Bond angles	N(1)–Ag(1)–O(1)	102.69(8)	N(1)–Ag(1)–C(11)#1	113.53(10)
	O(2)–Ag(1)–C(11)#1	143.64(10)	N(1)–Ag(1)–C(10)#1	145.80(9)
	O(2)–Ag(1)–C(10)#1	110.73(9)	C(11)#1–Ag(1)–C(10)#1	32.96(11)
	C(7)–N(1)–Ag(1)	129.61(17)	C(6)–N(1)–Ag(1)	124.87(16)
	C(15)–O(1)–Ag(1)	104.71(19)	C(11)–C(10)–Ag(1)#2	71.30(16)
	C(9)–C(10)–Ag(1)#2	108.66(17)	C(10)–C(11)–Ag(1)#2	75.74(17)
	C(8)–O(3)–C(8)#3	114.4(3)		

^a Symmetry transformations used to generate equivalent atoms: #1: *x*, *y* − 1, *z*; #2: *x*, *y* + 1, *z*; #3: −*x* + 1, −*z* + 1/2.

Etobb (64% yield). Mp: 101–103 °C. ^1H NMR (400 MHz, CDCl_3): δ : 7.28 (m, 4H, Ph), 4.12 (s, 2H, CH_2), 4.91 (s, 2H, OCH_2), 1.49 (s, 3H, CH_3). FT-IR (KBr ν/cm^{-1}): 756 $\nu_{(\text{O}-\text{Ar})}$, 1083 $\nu_{(\text{C}-\text{O})}$, 1475 $\nu_{(\text{C}=\text{N})}$, 1612 $\nu_{(\text{C}=\text{C})}$. UV-vis (DMF): λ = 280, 288 nm. Anal. calcd (%): C, 71.8; H, 6.6; N, 16.8; found (%): C, 71.4; H, 6.5; N, 16.6. A_M (DMF, 297 K): 0.76 $\text{S cm}^2 \text{mol}^{-1}$.

Bobb (75% yield). Mp: 177–178 °C. ^1H NMR (400 MHz, CDCl_3): δ : 7.78–7.80 (m, 4H), 7.17–7.30 (m, 10H), 6.94–6.96 (m, 4H), 5.29 (s, 4H), 4.80–4.82 (m, 4H). FT-IR (KBr ν/cm^{-1}): 1078 $\nu_{(\text{C}-\text{O}-\text{C})}$, 1496 $\nu_{(\text{C}=\text{N})}$, 1463 $\nu_{(\text{C}=\text{N}-\text{C}=\text{C})}$. UV-vis (DMF): λ = 279, 287 nm. Anal. calcd (%): C, 78.6; H, 5.7; N, 12.2; found (%): C, 78.5; H, 5.7; N, 12.3. A_M (DMF, 297 K): 1.13 $\text{S cm}^2 \text{mol}^{-1}$.

Aobb (53% yield). Mp: 69–73 °C. ^1H NMR (400 MHz, CDCl_3): δ : 7.75–7.79 (m, 2H, Ph-H), 7.25–7.30 (m, 6H, Ph-H), 5.81–5.99 (m, 2H, CH_2), 5.13–5.18 (m, 2H, CH_2), 5.00 (m, H, CH), 4.82–4.91 (s, 4H, CH_2). FT IR (KBr ν/cm^{-1}): 750 $\nu_{(\text{O}-\text{Ar})}$, 1066 $\nu_{(\text{C}-\text{O})}$, 1463 $\nu_{(\text{C}=\text{N})}$, 1612 $\nu_{(\text{C}=\text{C})}$. UV-vis (DMF): λ = 280, 288 nm. Anal. calcd (%): C, 73.7; H, 6.2; N, 15; found (%): C, 73.6; H, 6.2; N, 15.7. A_M (DMF, 297 K): 1.1 $\text{S cm}^2 \text{mol}^{-1}$.

Preparation of complexes

Three complexes were prepared using a similar procedure. An ethanol solution (5 mL) of sodium methacrylate (0.0217 g, 0.20 mmol) was added to an ethanol solution (3 mL) of AgNO_3 (0.0340 g, 0.20 mmol) and stirred for several minutes until the precipitate was generated. This was followed by the addition of ligand (Etobb: 0.0668 g, 0.20 mmol; Bobb: 0.0916 g, 0.20 mmol; Aobb: 0.0716 g, 0.20 mmol) in ethanol (5 mL). Thereafter, a few drops of acetonitrile were added and stirred at room temperature for 4 h until the white precipitate disappeared. Then, the colorless filtrate was slowly evaporated in the absence of light at room temperature. Crystals suitable for X-ray measurement were obtained after 3–5 weeks.

Complex 1 (67% yield). Mp: 112–114 °C (decomp.). Anal. calcd for $\text{C}_{52}\text{H}_{60}\text{Ag}_2\text{N}_{10}\text{O}_6$ (%): C, 54.9; H, 5.3; N, 12.3. Found (%): C, 54.8; H, 5.4; N, 12.3. FT IR (KBr ν/cm^{-1}): 746 $\nu_{(\text{O}-\text{Ar})}$, 1043 $\nu_{(\text{C}-\text{O})}$, 1552 $\nu_{(\text{C}=\text{N})}$. UV-vis (DMF): λ = 280, 287 nm. A_M (DMF, 297 K): 16.1 $\text{S cm}^2 \text{mol}^{-1}$.

Complex 2 (74% yield). Mp: 278–280 °C (decomp.). Anal. calcd for $\text{C}_{34}\text{H}_{31}\text{AgN}_4\text{O}_3$ (%): C, 62.7; H, 4.8; N, 8.6. Found (%): C, 62.8; H, 4.8; N, 8.7. FT IR (KBr ν/cm^{-1}): 756 $\nu_{(\text{O}-\text{Ar})}$, 1066 $\nu_{(\text{C}-\text{O})}$, 1552 $\nu_{(\text{C}=\text{N})}$. UV-vis (DMF): λ = 279, 287 nm. A_M (DMF, 297 K): 9.0 $\text{S cm}^2 \text{mol}^{-1}$.

Complex 3 (69% yield). Mp: 269–272 °C (decomp.). Anal. calcd for $\text{C}_{30}\text{H}_{32}\text{Ag}_2\text{N}_4\text{O}_5$ (%): C, 48.4; H, 4.3; N, 7.5. Found (%): C, 48.4; H, 4.4; N, 7.4. FT IR (KBr ν/cm^{-1}): 736 $\nu_{(\text{O}-\text{Ar})}$, 1053 $\nu_{(\text{C}-\text{O})}$, 1558 $\nu_{(\text{C}=\text{N})}$. UV-vis (DMF): λ = 279, 287 nm. A_M (DMF, 297 K): 9.5 $\text{S cm}^2 \text{mol}^{-1}$.

X-ray crystallography

For each complex, a suitable single crystal was mounted on a glass fiber, and the intensity data were collected on a Bruker APEX-II CCD diffractometer with graphite-monochromatized Mo $K\alpha$ radiation (λ = 0.71073 Å) at 296(2) K. Data reduction and cell refinement were performed using the SAINT suite of programs.⁶⁰ The absorption corrections were made by empirical

methods. The structures were solved by Direct Methods and refined by full-matrix least-squares against F^2 using SHELXTL software.⁶¹ All H atoms were found in different electron maps and subsequently refined in a riding model approximation with C–H distances ranging from 0.95 to 0.99 Å and $U_{\text{iso}}(\text{H}) = 1.2U_{\text{eq}}(\text{C})$ or $1.5U_{\text{eq}}(\text{C}_{\text{methyl}})$. The crystal data and experimental parameters relevant to the structure determination are listed in Table 2. Selected bond distances and angles are shown in Table 3.

Acknowledgements

The present research was supported by the National Natural Science Foundation of China (Grant No. 21367017), Natural Science Foundation of Gansu Province (Grant No. 1212RJZA037).

References

- 1 K. D. Mjos and C. Orvig, *Chem. Rev.*, 2014, **114**, 4540–4563.
- 2 C. Ceresa, A. Bravin, G. Cavaletti, M. Pellei and C. Santini, *Curr. Med. Chem.*, 2014, **21**, 2237–2265.
- 3 N. P. E. Barry and P. J. Sadler, *Chem. Commun.*, 2013, **495**, 106–5131.
- 4 C. Santini, M. Pellei, G. Papini, B. Morresi, R. Galassi, S. Ricci, F. Tisato, M. Porchia, M. P. Rigobello, V. Gandin and C. Marzano, *J. Inorg. Biochem.*, 2011, **105**, 232–240.
- 5 J. J. Liu, P. Galettis, A. Farr, L. Maharaj, H. Samarasinha, A. C. McGeachan, B. C. Baguley, R. J. Bowen, S. J. Berners-Price and M. J. McKeage, *J. Inorg. Biochem.*, 2008, **102**, 303–310.
- 6 S. J. Berners-Price, R. J. Bowen, P. Galettis, P. C. Healy and M. J. McKeage, *Coord. Chem. Rev.*, 1999, **185–186**, 823–836.
- 7 M. J. McKeage, P. Papathanasiou, G. Salem, A. Sjaarda, G. F. Swiegers, P. Waring and S. B. Wild, *Met.-Based Drugs*, 1998, **5**, 217–223.
- 8 P. Papathanasiou, G. Salem, P. Waring and A. C. Willis, *J. Chem. Soc., Dalton Trans.*, 1997, 3435–3443.
- 9 R. A. Haque, M. Z. Ghadhyeb, S. Budagumpi, A. W. Salman, M. B. Khadeer Ahamed and A. M. S. A. Majid, *Inorg. Chim. Acta*, 2013, **394**, 519–525.
- 10 T. Fuente, M. Martin-Fontecha, J. Sallander, B. Benhamu, M. Campillo, R. A. Medina, L. P. Pellissier, S. Claeyens, A. Dumuis, L. Pardo and M. L. Lopez-Rodriguez, *J. Med. Chem.*, 2010, **53**, 1357–1369.
- 11 O. Dogan, S. Demir, I. Ozdemir and B. Cetinkaya, *Appl. Organomet. Chem.*, 2011, **25**, 163–167.
- 12 T. Sun and K. Seff, *Chem. Rev.*, 1994, **94**, 857–870.
- 13 I. Rabin and W. Schulze, *J. Phys. Chem. B*, 2004, **108**, 14575–14578.
- 14 C. A. Tsipis, E. E. Karagiannis, P. F. Kladou and A. C. Tsipis, *J. Am. Chem. Soc.*, 2004, **126**, 12916–12929.
- 15 A. N. Khlobystov, A. J. Blake, N. R. Champness, D. A. Lemenovskii, A. G. Majouga, N. V. Zyk and M. Schroder, *Coord. Chem. Rev.*, 2001, **222**, 155–192.
- 16 D. J. Eisler and R. J. Puddephatt, *Inorg. Chem.*, 2006, **45**, 7295–7305.

- 17 H. J. Hao, D. Sun, Y. H. Li, F. J. Liu, R. B. Huang and L. S. Zheng, *Cryst. Growth Des.*, 2011, **11**, 3564–3578.
- 18 M. Sanchez-Moreno, E. Entrala, D. Janssen, J. M. Salas-Peregrin and A. Osuna, *Pharmacology*, 1996, **52**, 61–68.
- 19 W. J. Geary, *Coord. Chem. Rev.*, 1971, **7**, 81–122.
- 20 W. K. Dong, Y. X. Sun, G. H. Liu, L. Li, X. Y. Dong and X. H. Gao, *Z. Anorg. Allg. Chem.*, 2012, **638**, 1370–1377.
- 21 L. K. Thompson, B. S. Ramaswamy and E. A. Seymour, *Can. J. Chem.*, 1977, **55**, 878–888.
- 22 H. L. Wu, J. K. Yuan, Y. Bai, G. L. Pan, H. Wang, J. Kong, X. Y. Fan and H. M. Liu, *Dalton Trans.*, 2012, **41**, 8829–8838.
- 23 C. Buron, L. Stelzig, O. Guerret, H. Gornitzka, V. Romanenko and G. Bertrand, *J. Organomet. Chem.*, 2002, **664**, 70–76.
- 24 A. W. Salman, R. A. Haque and S. Budagumpi, *Polyhedron*, 2012, **42**, 18–23.
- 25 R. A. Haque, M. Z. Ghdhayeb, S. Budagumpi, A. W. Salman, M. B. Khadeer Ahamed and A. M. S. A. Majid, *Inorg. Chim. Acta*, 2013, **394**, 519–525.
- 26 M. Naodovic and H. Yamamoto, *Chem. Rev.*, 2008, **108**, 3132–3148.
- 27 I. Sloufovaa and M. Sloufb, *Acta Crystallogr., Sect. C: Cryst. Struct. Commun.*, 2000, **56**, 1312–1313.
- 28 T. S. Thakur and G. R. Desiraju, *Chem. Commun.*, 2006, 552–554.
- 29 C. S. Liu, P. Q. Chen, E. C. Yang, J. L. Tian, X. H. Bu, Z. M. Li, H. W. Sun and Z. Y. Lin, *Inorg. Chem.*, 2006, **45**, 5812–5821.
- 30 J. W. Steed, K. Johnson, C. Legido and P. C. Junk, *Polyhedron*, 2003, **22**, 769–774.
- 31 J. G. Liu, Q. L. Zhang, X. F. Shi and L. N. Ji, *Inorg. Chem.*, 2001, **40**, 5045–5050.
- 32 R. Olar, M. Badea, D. Marinescu, C. M. Chifiriuc and A. Finaru, *Eur. J. Med. Chem.*, 2010, **45**, 2868–2875.
- 33 H. F. Wang, R. Shen and N. Tang, *Eur. J. Med. Chem.*, 2009, **44**, 4509–4515.
- 34 A. Tarushi, G. Psomas, V. Psycharis and D. P. Kessissoglou, *Polyhedron*, 2009, **28**, 3272–3278.
- 35 S. Mukherjee, C. Basu, S. Chowdhury, A. P. Chattopadhyay, A. Ghorai, U. Ghosh and H. Stoeckli-Evans, *Inorg. Chim. Acta*, 2010, **363**, 2752–2761.
- 36 Q. Wang, Z. Y. Yang, G. F. Qi and D. D. Qin, *Eur. J. Med. Chem.*, 2009, **44**, 2425–2433.
- 37 H. L. Wu, J. K. Yuan, G. L. Pan, Y. H. Zhang, X. L. Wang, F. R. Shi and X. Y. Fan, *J. Photochem. Photobiol., B*, 2013, **122**, 37–44.
- 38 T. Ihara, S. Sueda, A. Inenaga, R. Fukuda and M. Takagi, *Supramol. Chem.*, 1997, **8**, 93–111.
- 39 R. Indumathy, T. Weyhermüller and B. U. Nair, *Dalton Trans.*, 2010, **39**, 2087–2097.
- 40 H. L. Wu, J. K. Yuan, Y. Bai, F. Jia, B. Liu, F. Kou and J. Kong, *Transition Met. Chem.*, 2011, **36**, 819–827.
- 41 H. L. Wu, J. K. Yuan, Y. Bai, G. L. Pan, H. Wang and X. B. Shu, *J. Photochem. Photobiol., B*, 2012, **107**, 65–72.
- 42 R. F. Pasternack, M. Caccam, B. Keogh, T. A. Stephenson, A. P. Williams and F. J. Gibbs, *J. Am. Chem. Soc.*, 1991, **113**, 6835–6840.
- 43 H. Torigoe, I. Okamoto, T. Dairaku, Y. Tanaka, A. Ono and T. Kozasa, *Biochimie*, 2012, **94**, 2431–2440.
- 44 H. Arakawa, J. F. Neault and H. A. Tajmir-Riahi, *Biophys. J.*, 2001, **81**, 1580–1587.
- 45 D. E. DiRico Jr, P. B. Keller and K. A. Hartman, *Nucleic Acids Res.*, 1985, **13**, 251–260.
- 46 S. Mahadevan and M. Palaniandavar, *Inorg. Chem.*, 1998, **37**, 693–700.
- 47 S. Satyanarayana, J. C. Dabroniak and J. B. Chaires, *Biochemistry*, 1992, **31**, 9319–9324.
- 48 S. Yellappa, J. Seetharamappa, L. M. Rogers, R. Chitta, R. P. Singhal and F. D'Souza, *Bioconjugate Chem.*, 2006, **17**, 1418–1425.
- 49 M. Shakir, M. Azam, M. F. Ullah and S. M. Hadi, *J. Photochem. Photobiol., B*, 2011, **104**, 449–456.
- 50 Z. Y. Yang, Y. Wang and Y. Wang, *Bioorg. Med. Chem. Lett.*, 2007, **17**, 2096–2101.
- 51 G. L. Pan, Y. C. Bai, H. Wang, J. Kong, F. R. Shi, Y. H. Zhang, X. L. Wang and H. L. Wu, *Z. Naturforsch., B: J. Chem. Sci.*, 2013, **68**, 257.
- 52 P. X. Xi, Z. H. Xu, X. H. Liu, F. H. Chen, Z. Z. Zeng, X. W. Zhang and Y. Liu, *J. Fluoresc.*, 2009, **19**, 63–72.
- 53 G. F. Qi, Z. Y. Yang, D. D. Qin, B. D. Wang and T. R. Li, *Chem. Pharm. Bull.*, 2008, **56**, 452–456.
- 54 J. R. Lakowicz and G. Webber, *Biochemistry*, 1973, **12**, 4161–4170.
- 55 C. P. Tan, J. Liu, L. M. Chen, S. Shi and L. N. Ji, *J. Inorg. Biochem.*, 2008, **102**, 1644–1653.
- 56 Z. Y. Guo, R. E. Xue, S. Liu, H. H. Yu, P. B. Wang, C. P. Li and P. C. Li, *Bioorg. Med. Chem. Lett.*, 2005, **15**, 4600–4603.
- 57 H. L. Wu, F. R. Shi, X. L. Wang, Y. H. Zhang, Y. C. Bai, J. Kong and C. P. Wang, *Transition Met. Chem.*, 2014, **39**, 261–270.
- 58 H. L. Wu, R. R. Yun, K. T. Wang, K. Li, X. C. Huang, T. Sun and Y. Y. Wang, *Z. Anorg. Allg. Chem.*, 2010, **636**, 1397–1400.
- 59 H. L. Wu, F. Kou, F. Jia, B. Liu, J. K. Yuan and Y. Bai, *Z. Anorg. Allg. Chem.*, 2012, **638**, 443–450.
- 60 Bruker, *APEX2 and SAINT*, Bruker Axs, Inc, Madison, WI, USA, 2007.
- 61 G. M. Sheldrick, *SHELXTL*, Siemens analytical X-ray instruments, Inc., Madison, WI, USA, 1996.